

24CS31 — 3 Hour Plan

Do these twelve things, in this order. Nothing else.

#	Study this	Minutes	Clock	Marks so far
1	Unitary / Hermitian check	8	0:08	5
2	Prove T is linear + find images	12	0:20	12
3	Least-squares solution	20	0:40	22
4	t^n -multiplication proof	15	0:55	29
5	Heaviside piecewise + LT	20	1:15	36
6	$L\{\sin \sqrt{t}\} \rightarrow L\{\cos \sqrt{t}/\sqrt{t}\}$	12	1:27	40
7	The Unit I definition	5	1:32	42
8	Solve ODEs by Laplace	22	1:54	49
9	SVD	22	2:16	54  you pass here
10	Orthogonal diagonalization	25	2:41	59
11	Span / basis / independence	12	2:53	64
12	Rotation-operator proof	12	3:05	65

What “marks so far” means.

If you stop studying at that row, that is what you score on the hardest of the four past papers. You need 50. You get there at step 9.

If you run out of time, do these next, in this order:

13. Positive-definite check — 8 min — makes Unit V a full 20
 14. Four fundamental subspaces — 15 min — adds 10 in Unit IV
 15. Complete solution of $Ax = b$ — 15 min — makes Unit IV a full 20
 16. Composition of transformations — 12 min — backup for Unit III
- All four take you to **72**.

Do not open these. QR factorization · convolution · periodic functions · improper integrals · Gram-Schmidt · Markov · quadratic forms · PCA · diagonalize and find A^n · orthogonal projection. They are long, and the numbers change every paper.

In the exam.

Read both questions in a unit. Tick the parts you can do. Answer the one with more ticks. Write every part you know. Never leave a unit blank.

UNIT I — Laplace Transforms

Answer Q1 → you get 13 of 20

Look for the question with the **t^n proof** in it. That is the one to answer.

In every past paper, the $\sin \sqrt{t}$ question is in the same one.

Q1a Definition — 2 marks

“Write the Laplace transform of a periodic function” or “Define the Laplace transform”

Just write the formula.

- Periodic, period T : $\frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$
- Definition: $L\{f\} = \int_0^\infty e^{-st} f(t) dt$

Q1b Show the $\cos \sqrt{t}$ result — 4 marks

Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2s^{3/2}} e^{-1/4s}$,

show $L\left\{\frac{\cos \sqrt{t}}{\sqrt{t}}\right\} = \frac{\sqrt{\pi}}{s^{1/2}} e^{-1/4s}$

Three lines.

- $\frac{d}{dt}(\sin \sqrt{t}) = \frac{\cos \sqrt{t}}{2\sqrt{t}}$
- So what you want is **$2 \times L\{\text{that derivative}\}$**
- Use $L\{f\} = sF(s) - f(0)$, and $f(0) = 0$

Q1c Prove the t^n rule — 7 marks

Prove $L\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \{F(s)\}$

Learn this as a script. It has been asked word-for-word in all four papers.

- Start: $F(s) = \int_0^\infty e^{-st} f(t) dt$
- Differentiate under the integral sign $\rightarrow F'(s) = -L\{t f(t)\}$
- Each further differentiation drops another $(-t)$
- Finish with **induction on n** . State the base case $n = 1$.

Q1d Long evaluation — SKIP — 7 marks

Leave this one. You have 13 of 20 without it.

UNIT II — Application of Laplace Transforms

Answer Q3 → you get 14 of 20

Look for the question with the **piecewise function** in it (the one with t^2 , $4t$, 8).

In every past paper, an **ODE question** sits right next to it.

Q3c Piecewise function + Laplace transform — 7 marks

Express $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$ using the Heaviside function, then find its Laplace transform.

Three steps.

- **Stack the jumps:** $f = t^2 + (4t - t^2)u(t-2) + (8-4t)u(t-4)$
- **Rewrite each bracket** in terms of $(t-a) \rightarrow$ you get $4 - \tau^2$ and $-8 - 4\tau$
- **Apply second shifting:** $L\{g(t-a)u(t-a)\} = e^{-as}G(s)$

Answer: $2/s^3 + e^{-2s}(4/s - 2/s^3) + e^{-4s}(-8/s - 4/s^2)$

Q3d Solve an ODE by Laplace transforms — 7 marks

It will be **one of these three**:

- $\frac{dx}{dt} - 2y = \cos 2t$ and $\frac{dy}{dt} + 2x = \sin 2t$, $x(0)=1$, $y(0)=0$
- $y'' + 4y' + 3y = e^{-t}$
- A circuit: $L \frac{di}{dt} + Ri = Ee^{-at}$

Same method for all three.

- Transform every term. Use $L\{y'\} = sY - y(0)$ and $L\{y''\} = s^2Y - sy(0) - y'(0)$
- Put the initial conditions in **at this step**. Then it is just algebra.
- One equation $\rightarrow$ solve for $Y(s)$, invert by partial fractions.
- Two equations $\rightarrow$ solve them like simultaneous equations in $X(s)$ and $Y(s)$, then invert each.

Q3a, Q3b Definition and short inverse — SKIP unless you have spare time

— 2 + 4 marks

These are overtime items 17 and 16. If you get them, Q3 becomes a full 20.

UNIT III — Vector Space and Linear Transformation

Answer Q5 → you get all 20

This is your best unit. No calculus anywhere.

Q5 has three parts and **you will have all three**.

Q5a Span / basis / independence — 6 marks

Is $\left\{ \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 4 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 1 & 2 \end{bmatrix} \right\}$ a basis of M_{22} ?

Or: do $\{(1,2,3), (-1,-1,0), (2,5,4)\}$ span R^3 ?

One method works for every version.

- Turn each object into a plain list of numbers. A 2×2 matrix becomes a list of 4.
- Stack those lists as the rows of a square matrix.
- Take the **determinant**.
- Not zero → independent → it is a basis / it spans. Zero → dependent.

Q5b Prove T is linear, then find images — 7 marks

(a) $T(x,y) = (3x+y, 2y, x-y)$. Find the images of $(1, 2)$ and $(2, -5)$.

(b) $T(ax^2+bx+c) = (a+b)x^2+c$. Find the image of $5x^2+6x+1$.

To prove it is linear, show two things:

- $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
- $T(k\mathbf{u}) = k T(\mathbf{u})$

Then just substitute the vectors.

Answers: $(5, 4, -1) \cdot (1, -10, 7) \cdot 11x^2 + 1$

Q5c Rotation proof — 7 marks

Show that $T(\mathbf{x}) = A\mathbf{x}$ rotates $\mathbf{x}$ through an angle θ about the origin, where

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Bookwork. Twelve minutes. No calculus.

- Write $\mathbf{x}$ in polar form: $\mathbf{x} = (r\cos\varphi, r\sin\varphi)$
- Multiply out $A\mathbf{x}$
- Use the compound-angle formulae → you get $(r\cos(\theta+\varphi), r\sin(\theta+\varphi))$
- Same length r , angle increased by θ . That is a rotation. Done.

UNIT IV — Orthogonal Projections

Answer Q8 → you get 10 of 20

Your weakest unit, on purpose. You take one 10-mark part and move on.

Do not attempt QR factorization — it is long and the matrix changes every paper.

Q8b Least-squares solution — 10 marks

Find the least-squares solution of $AX = b$, where

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 3 \\ 5 \\ 7 \\ -3 \end{bmatrix}$$

The whole question is one equation: $A^T A x = A^T b$

- $A^T A = \begin{bmatrix} 4 & 8 & 10 \\ 8 & 20 & 26 \\ 10 & 26 & 38 \end{bmatrix}$

- $A^T b = \begin{bmatrix} 12 \\ 12 \\ 20 \end{bmatrix}$

- Solve it → $\hat{x} = (10, -6, 2)$

- Substitute back to check. Thirty seconds, and it confirms all 10 marks.

This exact matrix and vector appeared twice, including in the most recent paper.

UNIT V — Applications of Eigenvalue Decomposition

Answer Q10 → you get 15 of 20

Look for the question with “**orthogonally diagonalize**” in it.

The best-value unit on the paper — the two 5-mark parts take two minutes each.

Q10b Unitary or Hermitian check — 5 marks

Is $A = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$ unitary?

Or: is $\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix}$ Hermitian?

Two minutes.

- Write A^* — transpose it, then flip the sign of every i .
- **Hermitian** if $A^* = A$
- **Unitary** if $A^*A = I$
- **Both of these say YES.** Show A^* written out in full.

Q10c Orthogonally diagonalize — 10 marks

Orthogonally diagonalize $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

This exact matrix appeared in 3 of the 4 papers. You already know the answer — practise producing it.

- Eigenvalues: **2, 3, 6**
- Eigenvectors: $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$
- All three eigenvalues are different, so the eigenvectors are **already orthogonal** — **no Gram-Schmidt needed**. Say this in your answer, it earns marks.
- Divide them by $\sqrt{2}, \sqrt{3}, \sqrt{6}$. Those are the columns of P .
- Then **$P^TAP = \text{diag}(2, 3, 6)$**
- Show the three dot products are 0.

Q9c SVD — this is in the OTHER question — 10 marks

Find the SVD of $A = \begin{bmatrix} 1 & 1 \\ 3 & -3 \end{bmatrix}$

No calculus — it is eigenvectors, twice. Do it as insurance in case Q10 looks wrong.

- $A^TA = \begin{bmatrix} 10 & -8 \\ -8 & 10 \end{bmatrix}$
- Eigenvalues **18** and **2** → singular values **$3\sqrt{2}$** and **$\sqrt{2}$**
- Its eigenvectors $(1, -1)/\sqrt{2}$ and $(1, 1)/\sqrt{2}$ are the columns of **V**
- Then **$u_i = A v_i / \sigma_i$** gives the columns of **U**
- Write **$A = U\Sigma V^T$**

Q10a Positive-definite check — overtime item 13 — 5 marks

Is $A = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$ positive definite?

Eight minutes, and it makes this unit a full 20.

- Work out the three leading minors: they are **1, 0, 0**
- So the answer is **NO — it is not positive definite**
- It is positive **semi**-definite. **Write that word** — just saying “no” loses marks.